MIL-D-46284 (Ord)
26 September 1960

MILITARY SPECIFICATION

DETONATOR, STAB, M45
LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of stab detonator designated as Detonator, M45.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-G-2550 - General Specification for Ammunition Except Small Arms Ammunition.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Inspection Terms and Definitions.

DRAWINGS

ORDNANCE CORPS

8797792 - Detonator, M45, Loading Assembly.
7548103 - Box, Set-up, Packing, Ammunition, Outer for Boosters, Detonators or Primers, Assembly, Details, Packing and Marking.
7548104 - Box, Set-up, Packing, Ammunition, Inner for Boosters, Detonators or Primers, Assembly, Details, Packing and Marking.
7548099 - Box, Packing, Ammunition for, Detonators.

FSC 1375

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PUBLICATIONS

ORDNANCE CORPS

ORD-M608-11 - Procedures and Tables for Continuous Sampling by Attributes.
IEL 7152281 - Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 71-78 - Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Interstate Commerce Commission Regulations are now a part of the Code of Federal Regulations (1949 Edition and revisions) available from the Superintendent of Documents, Government Printing Office, Washington 25, D.C. Orders for the above publication should cite "49 CFR 71-78 (latest revision)".

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Detonators.-The detonators shall comply with all requirements specified on Drawing (dwg.) 8797792, and with all requirements specified in applicable specifications.

3.3 Moisture content of lead azide.-The moisture content of the lead azide used in loading the detonators shall not exceed 0.50 percent when tested as specified in 4.3.2.

3.4 Moisture content of primer mixture.-The moisture content of the primer mixture used in loading the detonators shall not exceed 0.30 percent when tested as specified in 4.3.3.1.

3.5 Functioning.

3.5.1 Sensitivity.-The detonators shall function on the first attempt and shall cause perforation of the lead disc to a diameter of not less than 1/4 inch when tested as specified in 4.3.4.1.

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3.5.2 Waterproofness.-The detonators shall be waterproof as evidenced by normal functioning after being submerged under a minimum of 2 inches of water for not less than 48 hours when tested as specified in 4.3.4.2.

3.6 Workmanship.-All parts shall be free of chips, dirt, grease, rust, and other foreign material. The cleaning method used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-The supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define the terms used herein. Inspection shall be performed in accordance with this specification and other specifications referenced in any of the contractual documents.

4.1.1 Contractor quality assurance system.-The contractor shall provide and maintain an effective quality assurance system acceptable to the Government covering the supplies procured under this specification and shall perform all inspections required by the system. A current written description of the system shall be submitted to the contracting officer, and meet this approval, prior to initiation of production. It shall include a description covering controls for lot formation and identification, inspections to be performed, inspection stations, sampling procedures, methods of inspection (gages, measuring, and testing equipment), and provisions for control and disposition of non-conforming material. The written description will be considered acceptable when, as a minimum, it provides the quality assurance required by the provisions of 4.2 and 4.3 and the other documents referenced herein. Except in the case of 4.3.4.1 the contractor shall not be restricted to the inspection station nor the method of inspection listed in this specification provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether certain procedures of the contractor's system provide equal assurance, the comparable procedure of this specification shall apply. The contractor shall notify the Government of, and obtain approval for, any change tothe written procedure that affects the degree of assurance required by this specification or other documents referenced herein. In the case of the sensitivity test the weight of the ball and height of drop shall not differ from the values specified in 4.3.4.1 and on the applicable test equipment drawing.

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4.1.2 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.
b. Number of units of product inspected.
c. Results obtained for all inspection performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Certificates of analyses on all material procured directly by the contractor.
f. Number of items in the lot.
g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification shall consist of: (a) surveillance of contractor operations to determine that the methods and procedures of the written procedure are being properly applied and (b) Government product inspection. The Government inspector shall also ascertain prior to acceptance that the product complies with all the quality assurance provisions of other specifications referenced in any of the contractual documents.

4.1.3.1 Surveillance.-Surveillance shall include, but is not limited to:

a. Observation of procedures concerning lot formation and identification.
b. Observation of sampling procedures and application of acceptance criteria.
c. Determination that all required examinations and tests are performed in accordance with the prescribed procedures of this specification, or approved equivalents thereto.
d. Review of procedures for control and disposition of non-conforming material.

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4.1.3.2 Product inspection.-Physical inspection by the Government of contractor accepted product will be performed to the extent necessary to assure contractor compliance with the requirements of this specification and to determine the reliability of contractor records. Severity of Government inspection of individual characteristics will be directly related to the seriousness classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records, (Specification MIL-G-2550).

4.1.3.3 Deviations and deficiencies.-Deviations from the prescribed or agreed upon procedures, or instances of poor practice which might have an effect on the quality of the product, shall be called to the attention of the contractor immediately. To avoid interference with operation, the contractor shall designate a responsible official, or officials, to whom the Government inspector will report such instances. Failure of the contractor to correct deficiencies promptly shall be cause for suspension of acceptance until correction has been made or until conformance to prescribed criteria has been demonstrated.

4.2 Inspection provisions.

4.2.1 Lot formation.-A lot shall consist of detonators produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification, and process changes not affecting safety, performance, or fit, as determined by the Government, shall not necessitate changing the lot number. Each lot shall contain:

a. Parts bearing the same interfix lot number from one manufacturer.
b. RDX from not more than one lot.
c. Lead azide from not more than one lot.
d. Primer mixture from one batch or blending of batches which results in homogeneity of all primer mixture used in loading the entire detonator lot.

4.2.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Handbook ORD-M608-11 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Assembly, prior to loading (see dwg. 8797792).

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	AQL 0.25 percent	
101.	Coating compound or lacquer on exposed surface of disc	Visual

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Minor: AQL 0.40 percent

201.	Evidence of poor workmanship (see 3.6)	Visual

4.2.2.2 Assembly (see dwg. 8797792).

Categories	Defects	Method of inspection

Critical: None defined

Major: AQL 0.65 percent

101.	Length	Gage
102.	Diameter	Gage
103.	Color of lacquer incorrect (both ends)	Visual
104.	Not crimped full 360°	Visual
105.	Disc loose	Manual
106.	Disc punctured	Visual
107.	Cup damaged	Visual

Minor: AQL 0.40 percent

201.	Evidence of poor workmanship (see 3.6)	Visual

4.2.2.3 Box, set-up, packing, ammunition, inner (see dwg. 7548104).

Categories	Defects	Method of inspection

Critical: None defined

Major: AQL 0.25 percent

101.	Box damaged exposing contents	Visual
102.	Label missing or incorrect	Visual

Minor: None defined

4.2.2.4 Box, set-up, packing, ammunition, outer (see dwg. 7548103).

Categories	Defects	Method of inspection

Critical: None defined

Major: AQL 0.25 percent

101.	Barrier bag damaged	Visual
102.	Label missing or incorrect	Visual

Minor: None defined

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4.2.2.5 Box, packing, ammunition for detonators (see dwg. 7548099).

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	AQL 0.40 percent	
101.	Box damaged	Visual
102.	Nails missing or improperly driven	Visual
103.	Strapping missing or loose	Visual-Manual
104.	Marking incomplete or misleading	Visual
Minor:	None defined	

4.2.3 Testing.

4.2.3.1 Binder-lubricant content of RDX (see dwg. 8797792) - Major defect.- The binder-lubricant content of the RDX shall be determined for each lot at the time of loading. A sample of 5 grams shall be selected from each lot and tested as specified in 4.3.1. If the binder-lubricant content is in excess of the percentage specified and if loading has not begun, the lot of RDX shall be rejected until corrective action has been accomplished as verified by repeating the binder-lubricant determination specified in 4.3.1. If loading has begun, action on the unloaded RDX shall be as specified above and all detonators loaded with the questionable RDX shall be rejected.

4.2.3.2 Moisture content of lead azide (see 3.3) - Major defect.-The moisture content of the lead azide shall be determined for each lot at the time of loading. A sample of 2 grams shall be selected and tested as specified in 4.3.2. If the moisture content is in excess of the percentage specified and loading has not begun, the lot of lead azide shall be rejected until proper corrective action has been accomplished as verified by repeating the moisture content test specified in 4.3.2. If loading has begun, action on the unloaded lead azide shall be as specified above and all detonators loaded with the questionable lead azide shall be rejected.

4.2.3.3 Primer mixture.

4.2.3.3.1 Moisture content (see 3.4) - Major defect.-The moisture content of the primer mixture shall be determined for each lot at the time of loading. A sample of 0.7 gram shall be selected and tested as specified in 4.3.3.1. If the moisture content is in excess of the percentage specified, and if loading has not begun, the lot of primer mixture shall be rejected until proper corrective action has been accomplished as verified by repeating the moisture content test specified in 4.3.3.1. If loading has begun, action on the unloaded primer mixture shall be as specified above and all detonators loaded with the questionable primer mixture shall be rejected.

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4.2.3.3.2 Composition (see dwg. 8797792) - Major defect.-The composition of the primer mixture shall be determined for each lot at the time of loading. A sample of 4 grams shall be selected and tested as specified in 4.3.2.2. If the composition of the primer mixture fails to conform to the requirements specified in the applicable drawing, and if loading has not begun, the lot of primer mixture shall be rejected. If loading has begun, detonators loaded with the questionable primer mixture shall be rejected.

4.2.3.4 Functioning.

4.2.3.4.1 Sensitivity (see 3.5.1) - Major defect.-A sample of complete detonators shall be selected in accordance with Standard MIL-STD-105, using code letter P and an AQL of 0.15 percent. The sample detonators shall be tested as specified in 4.3.4.1, using test equipment specified in 4.2.4. Rejection shall be in accordance with Standard MIL-STD-105.

4.2.3.4.2 Waterproofness (see 3.5.2) - Major defect.-A sample of 25 detonators shall be selected from each lot and subjected to the test specified in 4.3.4.2, using test equipment specified in 4.2.4. If one or more detonators fail to function satisfactorily, as evidenced by testing as specified in 4.3.4.2, the lot shall be rejected.

4.2.4 Inspection equipment.-For the performance of the examinations and tests specified in 4.2 and 4.3, when applicable, the contractor shall have available, and utilize correctly, all necessary gages, measuring, and testing equipment identified by Index of Inspection Equipment Lists No. 7152281, or approved equivalents thereto. (see 4.1.1) When inspecting the contractor's inspection equipment, the Government inspector will determine that the contractor has available, and utilizes correctly, gages, measuring, and testing equipment of the required accuracy and precision and that the instruments are of the proper type and range to make measurements of the required accuracy. The contractor will have available a set of master gages, standards, and appropriate instruments for regularly scheduled calibration of his inspection equipment. Records of such regularly scheduled calibration will be maintained by the contractor and made available for review by the Government. The calibration of gages, standards, and instruments will be periodically checked by authorized Government personnel.

4.3 Test methods and procedures.

4.3.1 Determination of binder-lubricant content of HBX.-The calcium resinate, graphite and total binder-lubricant content of HBX shall be determined as follows:

4.3.1.1 Calcium resinate.-An accurately weighed portion of 5 grams of the sample shall be transferred to a tared medium porosity sintered-glass crucible. The sample shall be extracted with three 20 milliliters (ml.) portions of chloroform which have been previously saturated with HBX, maintaining each portion in contact with the residue for approximately 5 minutes before applying suction. Any lumps which may persist shall be broken up by stirring occasionally and the sides of the crucible washed down with each 20 ml. portion of chloroform to insure complete removal of the calcium resinate. The residue shall be aspirated until the odor of chloroform is no longer detectable; the crucible and contents

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shall be placed in an oven at 100 degrees centigrade (°C.) to 105°C. for 1/2 hour cooled in a desiccator and weighed. The above extraction shall be repeated, drying and weighing with individual 20 ml. portions of chloroform until the loss in weight on subsequent extractions totals less than 0.003 grams. The loss in weight of the curcible plus contents shall be calculated to percent calcium resinate in the sample on a moisture free basis as follows:

$$\text{percentage of Calcium Resinate} = \frac{100A}{W}$$

where:

A = loss in weight of the crucible plus contents.
W = weight of sample.

4.3.1.2 Graphite.-The residue remaining after the calcium resinate has been removed shall be extracted with two 20 ml. portions of dimethyl formamide, each portion maintained in contact with the residue for approximately 5 minutes with intermittent stirring before applying suction. The crucible and contents shall be washed with a minimum of 3 ml. portions of acetone, each portion maintained in contact with the residue for approximately one minute with continuous stirring before applying suction, until the following qualitative test for RDX in the acetone washings is negative. The acetone filtrate shall be transferred to a 25 ml. beaker, evaporated to dryness on a steam bath and several drops of a solution of chromotropic acid added in 36N sulfuric acid. The formation of a reddish-brown color as compared to a pink color obtained by a blank prepared in an identical manner indicates the presence of RDX. The residue shall be aspirated until the odor of acetone is no longer detectable; the crucible and contents shall be placed in a drying oven at 100°C. to 105°C. for 1/2 hour, cooled in a desiccator and weighed. The weight of the residue in the crucible shall be calculated to the percentage of graphite in the sample on a moisture-free basis as follows:

$$\text{percentage of Graphite} = \frac{100A}{W}$$

where:

A = increase in weight of filtering crucible.
W = weight of sample.

4.3.1.3 Total binder-lubricant.-The percentage of total binder-lubricant in the sample shall be calculated by adding the percentages of calcium resinate and graphite found.

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4.3.2 Determination of moisture content of lead azide.-A tared dried aluminum dish 2 inches in diameter shall be covered with a sheet of polytetrafluoroethylene 2 1/2 inches square and sufficiently flat to form a tight cover. An accurately weighed portion of approximately two grams of the sample shall be transferred to the moisture dish and the dish and the contents shall be transferred to an oven and dried at 95°C. ± 2°C. for 2 hours. The cover shall be placed in the sulfuric acid desiccator during the drying period. After drying, the dish shall be quickly transferred to the desiccator and allowed to cool to room temperature. The polytetrafluoroethylene cover shall be placed on the dish and the assembly weighed. The loss in weight shall be calculated to percentage of moisture in the lead azide to determine compliance with 3.3 as follows:

$$\text{percentage of moisture} = \frac{(A-B)}{A} \times 100$$

where:

A = the weight of the sample before drying.
B = the weight of the sample after drying.

4.3.3 Primer mixture.

4.3.3.1 Moisture content of primer mixture.-The specified sample shall be accurately weighed to the nearest milligram and dried to a constant weight at 60°C. ± 5°C. or in a desiccator containing calcium chloride. The loss in weight shall be calculated as the percentage of moisture.

4.3.3.2 Composition.

4.3.3.2.1 Potassium chlorate.

4.3.3.2.1.1 Preparation of solutions.

4.3.3.2.1.1.1 Solution No. 1.-To approximately 800 ml. of distilled water in a one liter amber glass reagent bottle, 40 gm. of potassium sulfocyanate shall be added. The bottle shall be sealed with a rubber stopper and the reagent bottle and contents shaken until the solid material is dissolved. To this solution 1.0 gm. of lead thiocyanate, 1.0 gm. of antimony sulfide and 1.0 gm of lead azide shall be added. The bottle shall again be stoppered and the reagent bottle and contents vigorously shaken for several minutes and then allowed to stand overnight. Portions of the saturated solution shall be filtered before using.

4.3.3.2.1.1.2 Solution No. 2.-To approximately 400 ml. of distilled water in a one-half liter amber glass reagent bottle, 1.0 gm. of lead thiocyanate, 1.0 gm. of antimony sulfide and 1.0 gm of lead azide shall be added. The bottle shall be sealed with a rubber stopper and the reagent bottle and contents vigorously shaken forseveral minutes and then allowed to stand overnight. Portions of the saturated solution shall be filtered before using.

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4.3.3.2.1.2 Procedure.-A portion of approximately 1.0 gm. of the sample shall be accurately weighed directly into a tared 30 ml. sintered glass crucible of medium porosity. From a burette 3.0 ml. of solution No. 1 shall be added to the crucible. The solution shall be swirled for exactly one minute, taking care not to spill any liquid; aspirated with suction and the bottom of the crucible wiped dry, using tissue paper. The washing shall be continued, using one more 3.0 ml. portion, two 2.0 ml. and two 1.0 ml. portions, making a total of 12.0 ml. The crucible shall be removed and the bottom of the crucible, the funnel, the adapter and the suction flask washed with water. The test tube shall be placed in the flask and the stem of the filtering funnel inserted into the test tube. The crucible and contents shall be washed with one 1.0 ml. portion of Solution No. 1 as above, and aspirated with suction, catching the filtrate in the test tube. The test tube shall be removed, 1.0 ml. of distilled water added and the chlorate ion, determined as follows:

The tube shall be inclined at an angle of about 45 degrees and from a dropper about 0.5 ml. of a solution containing 1.0 gm. of diphenylamine in 100 gm. of concentrated sulfuric acid shall be run down the side of the tube, so that there will be two distinct layers. If an appreciable amount of chlorate ion is present, a distinct blue ring will be visible at the border of the two layers. On shaking the tube, the ring disappears, but the contents assume a blue coloration, which lasts several seconds, depending on the amount of chlorate ion present. If only a trace of chlorate ion is present the blue ring may not form, but upon shaking the test tube, a slight bluish coloration will appear momentarily throughout the solution, lasting only a fraction of a second. If a definite blue ring forms, the washing shall be repeated as before using one 1.0 ml. portion of Solution No. 1, and finally a 0.5 ml. portion of the same solution, as needed, depending on the outcome of the tests for the chlorate ion. (Usually a total of 14.0 or 14.5 ml. of Solution No. 1 is sufficient). After complete removal of potassium chlorate the inside of the crucible shall be washed with 1.0 ml. of Solution No. 2 and aspirated immediately. The bottom of the crucible shall be wiped dry with tissue paper and dried in an oven at 80°C. ± 1°C. for 30 minutes, cooled in a desiccator and weighed. The percentage of potassium chlorate shall be calculated as follows:

$$\text{Potassium Chlorate, Percent} = \frac{B-(WM)\ \ 100}{W\ (1-M)}$$

where:

B = loss in weight of the contents of the crucible.
W = weight of the sample.
M = percent moisture expressed as a decimal fraction.

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4.3.3.2.2 Antimony Sulfide.

4.3.3.2.2.1 Preparation of solution of Ferric ammonium sulfate.-To approximately 120 ml. of distilled water, in a one-half liter reagent bottle, 125 gm. of ferric ammonium sulfate shall be added. The bottle shall be stoppered and shaken vigorously for several minutes and then allowed to stand overnight. Portions of the saturated solution shall be filtered before using.

4.3.3.2.2.2 Procedure.-A portion of approximately 1 gm. of the sample shall be accurately weighed directly into a tared 30 ml. sintered glass crucible of medium porosity. Five ml. of 1 percent nitric acid shall be added and the crucible constantly swirled for one minute in order to dissolve all the ingredients of the powder except antimony sulfide. Care shall be taken not to spill any liquid. The mixture shall be filtered by applying suction employing a vacuum flask. The crucible shall be removed from the adapter and the bottom wiped dry with tissue paper. The operations of washing, using 5.0 ml. of one percent nitric acid each time and swirling for exactly one minute shall be repeated. After a total of 40 ml. of one percent nitric acid has been used, the crucible shall be removed and the bottom of it washed with water. The funnel, the funnel adapter and the vacuum flask shall be washed with water; the filtrate and washings shall be discarded. About 1 ml. of a saturated solution of ferrec ammonium sulfate shall be placed in the flask. The contents of the crucible shall be treated as before with 5.0 ml. of one percent nitric acid and filtered by means of suction into the flask. If the liquid in the flask turns red, due to the presence of the sulfocyanate ion in the filtrate, the washing of the contents in the crucible shall be repeated with additional 5.0 ml. portions of one percent nitric acid until a negative test for the sulfocyanate ion is affected. (Usually a total of 50 ml. is sufficient). The crucible and contents shall be rinsed with alcohol and then with ether, and then aspirated until the odor of ether is no longer detectable. The crucible and contents shall be dried in an oven at 80°C. ± 1°C. for 30 minutes, cooled in a desiccator and weighed. The percentage of antimony sulfide shall be calculated as follows:

$$\text{Antimony sulfide, percent} = \frac{A \times 100}{B}$$

where:

A = weight of residue in the crucible.
B = weight of sample on a moisture-free basis.

4.3.3.2.3 Lead azide.-The lead azide content shall be determined using the apparatus shown assembled in Figure 1 and the temperature maintained at as near constant as is practicable. This shall be accomplished by protecting the gas system from the effects of air currents or bodies at appreciably different temperatures. A weighed portion of approximately 2 grams of the sample shall be transferred

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to a 125 ml. Erlenmeyer flask (D), and 5 ml. of 10-percent silver nitrate solution and 10 ml. of distilled water shall be added. The mixture shall be shaken and the sides of the flask washed down with distilled water. The delivery tube (A) shall be filled with 60-percent ceric ammonium nitrate solution. The Erlenmeyer flask (D) shall be connected tightly to the gas burette (E) (a gas burette calibrated in 0.1 ml. and having a total capacity of 100 ml. for the calibrated portion is satisfactory) by means of a tightly fitting rubber stopper (C), and this assembly clamped in position. The stopcock (F) shall be opened and, using the leveling bulb (G), the water level in the gas burette (E) shall be adjusted to the zero point. The stopcock (F) shall be closed and the system shall be tested for gas-tightness by raising and lowering the leveling bulb (G) and re-leveling the water in the bulb (G) with that in the burette (E). The system shall be considered to be gas-tight if the water level in the burette (E) returns to the zero point. Five ml. of 60-percent ceric ammonium nitrate solution shall be added from the delivery burette (A), the leveling bulb (G) shall be lowered to the vicinity of the 30 ml. mark on the gas burette (E) and the reaction flask (D) shaken very vigorously, until no further increase in gas volume occurs. (This may conveniently be accomplished by agitation of the entire assembly). Approximately 200 ml. of water shall be transferred to a 400 ml. beaker (H) at the temperature of the gas burette as shown by thermometer (I), and the beaker raised under the reaction flask (D) until the water level rises about 2 inches above the bottom of the flask. The beaker shall be clamped in position and the apparatus allowed to stand with occasional shaking until the temperature of the gas in reaction flask (D), as indicated by thermometer (B) is the same as that of the gas in the gas burette (E), as indicated by thermometer (I), and until no further change in gas volume occurs. The height of the leveling bulb (G) shall be adjusted so that the level of the water in it is the same as the level of the water in the gas burette (E). The observed volume in the gas burette (E), the temperature of the gas, and the atmospheric pressure shall be recorded. The percentage of lead azide in the sample shall be calculated on a moisture-free basis as follows:

$$\text{Lead azide, percent} = \frac{(A-5.0)\,(B-C)\,0.0570 \times 1.08}{W\,(100-D)\,(1 + 0.00367t)}$$

where:

A - gas burette reading, ml.

B - atmospheric pressure, mm. of mercury.

(Note.-If the pressure is observed on a mercurial barometer having a brass scale, the correct atmospheric pressure shall be calculated as follows:

Atmospheric pressure (B), mm. = E-0.000163EF

where:

E = reading in mm. of mercurial barometer.

F = temperature of barometer in °C.

C = vapor pressure of water at t °C.

D = percentage of moisture in the sample.

t = temperature of gas burette and reaction flask, °C.

W = weight of sample, grams.

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4.3.3.2.4 Lead thiocyanate.-The percentage of lead thiocyanate in the sample shall be calculated on a moisture-free basis by subtracting from 100 the sum of the percentages of potassium chlorate, antimony sulfide and lead azide.

4.3.4 Functioning.

4.3.4.1 Sensitivity.-The detonator shall be assembled to the applicable test fixture (see 4.2.4) so that the firing pin will strike the sensitive end of the detonator. A steel ball 1 3/16 inches in diameter and weighing 3.95 ounces maximum, shall be dropped on the firing pin, to function the detonator, from a height of 3 inches maximum. The height of drop shall be measured as the height from the top of the firing pin to the bottom of the ball. A detonator that fails to fire or fails to create a hole in the lead disc of the required size shall be classed defective.

4.3.4.2 Waterproofness.-The detonator shall be submerged in water to the depth specified for the time specified. Within one hour after removal from the water the detonator shall be subjected to the sensitivity test specified in 4.3.4.1. Any detonator that fails to fire or fails to create a hole in the lead disc of the required size shall cause the detonator to be classed defective.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Levels B and C (see 6.3).-The detonators shall be preserved, marked and packaged in accordance with dwgs. 7548103 and 7548104.

5.2 Packing.

5.2.1 Levels C.-The detonators shall be packed and marked in accordance with dwg. 7548099).

5.3 Data cards.-Data card information shall be as specified in Specification MIL-G-2550.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number, and date of this specification.

6.2 Check test for possible deterioration.-The following test is designed to prevent the loading of detonators which have deteriorated into a major ammunition item. The test should be performed on detonators from each lot which has been stored under normal conditions for more than two years or under adverse conditions for any period of time. If the specification for the ammunition item into which the detonators are loaded does not require that the check test be performed, the contracting officer should arrange to have the contractor reimbursed for any expenses incurred in the performance of the check test for possible deterioration.

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6.2.1 Sensitivity.-A sample of detonators shall be selected in accordance with Standard MIL-STD-105, using code letter P and and AQL of 0.25 percent. The sample detonators shall be tested as specified in 4.3.4.1, using test equipment specified in 4.2.4. Rejection shall be in accordance with Standard MIL-STD-105.

6.3 Level B, packaging.-In the event the expected storage time of the detonators is in excess of 180 days the barrier bag shall be used as specified on the applicable drawing. The expected storage time shall be determined by the procuring agency.

Notice.-When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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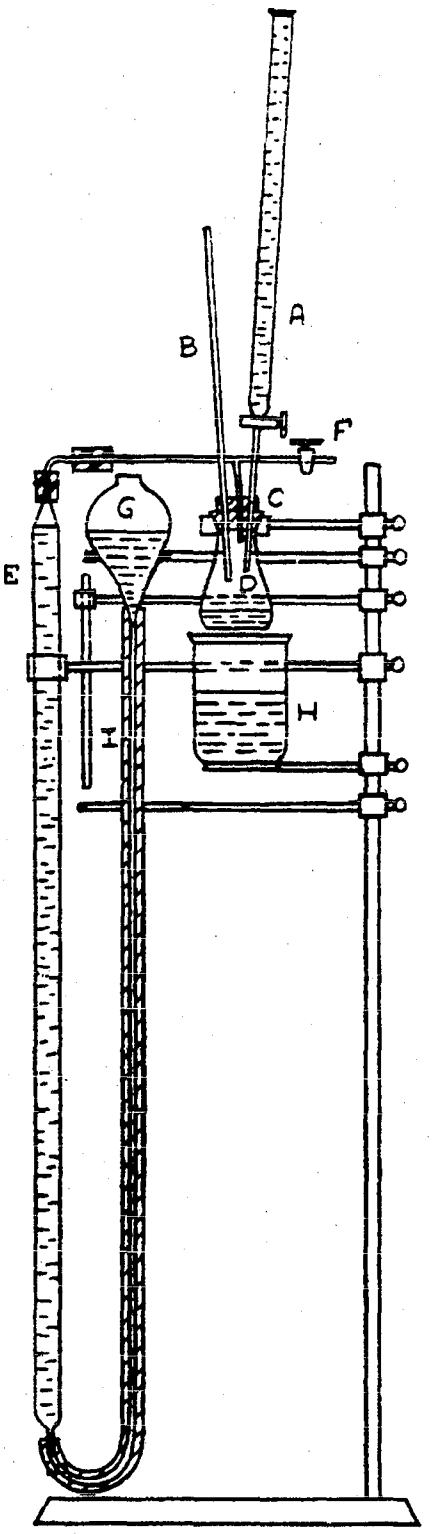


FIGURE 1

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